

answers are marked as positive, while others are negative. Contrastive learning is ideal since it allows us to identify positive KG paths and answer conversational questions. Besides path information, the entire dialog history, the verbalized answers, and domain information of the conversation are also used to help learn better path embeddings. Continuing along this trajectory, CornNet [35] advocates for the utilization of language models to generate additional reformulations. This approach aids in enhancing the model's comprehension of original, concise, and potentially challenging questions, ultimately leading to improved performance.

In [7], the authors employed reinforcement learning to train an agent that reformulates input questions to aid the system's understanding. In [20], an encoder-decoder model is used to transform natural language questions into logical queries for finding answers. In [24], a Transformer model is used to generate logical forms, and graph attention is introduced to identify entities in the query context. Other systems, such as Google's Lambda [56], Amazon Alexa [2], Apple's Siri, and OpenAI's ChatGPT, are also pursuing this task.

Question rewriting, which aims to reformulate an input question into a more salient representation, is also used in multi-turn question answering. This can improve the accuracy of search engine results or make a question more understandable for a natural language processing (NLP) system. In [58], a unidirectional Transformer decoder is proposed to automatically rewrite a user's input question to improve the performance of a conversational question answering system. In [14], authors propose a Seq2Seq model to rewrite the current question according to the conversational history and introduce a new dataset named CANARD. In [16], query rewriting rules are mined from a background KG, and a query rewriting operator is used to generate a new question.

6 LLM WITH KNOWLEDGE GRAPH REASONING

with the advent of ChatGPT, large language models have demonstrated great performance in many downstream tasks. Previously, we introduced knowledge graph reasoning. We know that knowledge graphs contain accurate structural knowledge and are very interpretable, however, most knowledge graphs are incomplete, lack language understanding. While language models are general knowledge, they are good at language understanding. However, they suffer from hallucination, lack interpretation, lacking new knowledge. By combining them together, we can build a model that is not only accurate but also interpretable.

When combining knowledge graphs with large language models, there are three different ways. The first category is letting LLMs enhance knowledge graph reasoning, where LLMs serve as a component in the reasoning process. The second category is Knowledge graph reasoning enhance LLMs where knowledge graph reasoning can be used to mitigate the LLM's hallucination problem. Finally, integrating knowledge graph reasoning with LLMs in a mutually beneficial way, so that they can help each other. Figure 5 shows the classification.

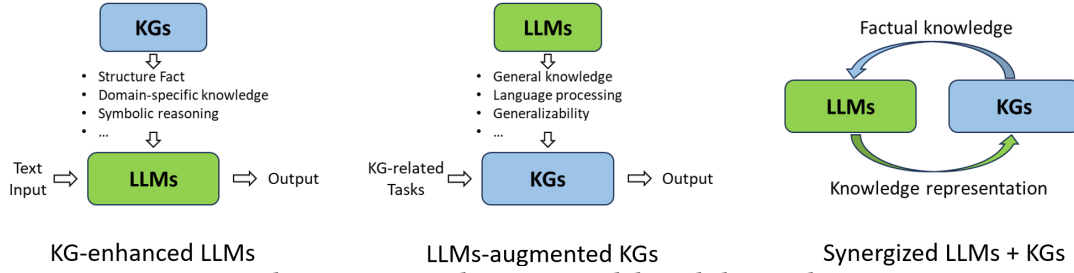
6.0.1 Knowledge graph enhances LLMs. Many methods have been proposed to utilize knowledge graph to boost the LLMs performance. In QA-GNN [73], it combines LLM and knowledge graph to

answer multi-choice questions. First of all, given a QA context, it will use the language model to encode question to a vector presentation, then use stand method to retrieve a knowledge graph subset, like linking entities and get their neighbors, and reasoning according to the subgraph. Then QA-GNN is based on two core ideas. First, in order to better identify which knowledge graph nodes relevant to current question, they propose language condition KG nodes scoring where they use a pretrained language model to compute the probability of each entity condition on the current question. Secondly, to jointly reason with language models, they connect the question text and kg to form a joint graph, which we call working graph, they mutually update their representations, through graph neural networks. Finally, they combine the representation of the language model and kg to predict the final answer. Following this direction, GreaseLM [81] merges LM with graph neural network by using a merging layer. To encode the knowledge graph structure information, multiple merging layers are used.

Now, we have introduced how to use knowledge graphs to help retrain language models to better answer different types of questions. However, when the size of the model becomes large, retrain or finetune the model will be very time consuming. An alternative way is to retrieval knowledge from external sources to help the language model generate correct answers. This approach allows for more targeted adjustments without the need for extensive retraining. This type of method is called Retrieval-Augmented Generation short for RAG.

In KG-GPT [27], the authors propose a method to utilize language models and knowledge graphs to answer more complex natural language questions. The idea is that a sentence of question, it uses llms to decompose the original sentence to several sub-sentences and find answers for each sub-sentence. And finally, find the answer for the whole sentence. In REPLUG [50], a new retrieval-augmented LM framework where the language model is viewed as a black box and the retrieval component is added as a tunable plug-and-play module. Given an input context, REPLUG first retrieves relevant documents from an external corpus. The retrieved documents are prepended to the input context and fed into the black-box LM to make the final prediction. Because the LM context length limits the number of documents that can be prepended, they also introduce a new ensemble scheme that encodes the retrieved documents in parallel with the same black-box LM. Then we pass the concatenation of each retrieved document with the input context through the LM in parallel, and ensemble the predicted probabilities. They have also developed REPLUG LSR. Instead of adjusting the language model to fit the retriever, they adapt the retriever to the language model. they train it to find documents that help improve the language model's understanding, while keeping the language model unchanged. The training process involves four steps: (1) finding and assessing the relevance of documents, (2) scoring these documents using the language model, (3) adjusting the retrieval model based on how different the retrieval and language model scores are, and (4) updating the index of the data in real-time.

Previous approaches of retrieval-augmented language models can only answer questions where answers are contained locally within regions of text and fail on answering global questions such as "what are the main themes in the dataset?" To solve these questions,



the work proposes a method called GraphRAG [13] which is a two-step process. The approach uses an LLM to build a graph-based text index in two stages: first to derive an entity knowledge graph from the source documents, then to pregenerate community summaries for all groups of closely-related entities. Given a question, each community summary is used to generate a partial response, before all partial responses are again summarized in a final response to the user.

6.0.2 LLMs enhances knowledge graph reasoning. Traditional methods retrieve information from KGs, augment the prompt accordingly, and feed the increased prompt into LLMs. In this paradigm, LLM plays the role of translator which transfers input questions to machine-understandable command for KG searching and reasoning, but it does not participate in the graph reasoning process directly. Its success depends heavily on the completeness and high quality of KG, which means that if the knowledge graph contains many missing edges, the answer won't be good. So, recently, some people tried to explore how to mitigate the knowledge graph incompleteness problem by treating the LLM as an agent to travel KGs and perform reasoning based on paths. In Think-on-Graph [53], LLM is treated as an agent to traverse on knowledge graph to find answers. Now, let's see an example, given question "What is the majority party now in the country where Canberra is located?" The llm identified the topic entity is Canberra, by checking its local neighbourhood, Australia has the highest score. But the LLM notices that there is not enough information to get the answer. So it travels to Australia and check its neighbors.

6.0.3 LLMs and knowledge graph reasoning mutually help each other. Finally, let us introduce the third part, how to let knowledge graph reasoning and large language models mutually help each other.

One of the most important tasks in knowledge graph reasoning is to learn the embedding for the KG entities. It has been shown that the learned embedding from KG can be used to improve the performance of Pre-trained Language Models. On the other hand, for each node in the knowledge graph, we can find many text descriptions to describe it. This text information can be used to learn the node embedding. The strong ability of pre-trained language model can help us learn the embedding. So, in KEPLER [62], they propose a model which can also solve two problems at the same time. In this paper, Roberta is chosen as the encoder. Given a triplet, each node has a text description. Roberta is used to learn entity embedding. And the learned embedding will be used to calculate the knowledge graph embedding loss. On the other hand, conventional

masked language model is used. It requires the model to predict the masked token within the text. The final loss is the summation of these two losses.

To leverage the structure information during the reasoning process, In [77], this work proposes to use a graph attention network to provide structure-aware entity embeddings for language modeling. More specifically, the language module produces contextual representations as initial embeddings for KG entities and relations given their descriptive text. Then, graph neural network is used to update the node embedding and relation embedding. Next, the learned knowledge graph embedding will be used as the input of the language model. And the output of the language model will be used to solve different downstream tasks. There are several advantages of this method. For example, the joint pre-training effectively projects entities/relations and text into a shared semantic latent space, which eases the semantic matching between them.

7 CONCLUSION AND FUTURE DIRECTIONS

7.1 Conclusion

Knowledge graphs serve as intuitive repositories of human knowledge, facilitating the inference of new information. However, traditional symbolic reasoning struggles with incomplete and noisy data often found in these graphs. Neural Symbolic AI, a fusion of deep learning and symbolic reasoning, offers a promising solution by combining the interpretability of symbolic methods with the robustness of neural approaches. Furthermore, the advent of large language models (LLMs) has opened new frontiers, enhancing knowledge graph reasoning capabilities. We explore these advancements by categorizing knowledge graph reasoning into four main areas: single-hop queries, complex logical queries, natural language questions and LLMs with knowledge graph reasoning. For single-hop queries, we review techniques that efficiently handle direct relationships between entities. Complex logical query reasoning involves multi-hop inference, where we discuss methods that manage intricate relationships and multiple steps of reasoning. Finally, we delve into reasoning for natural language questions, including both single-turn and multi-turn dialogues. This survey aims to provide a comprehensive overview, bridging the gap between different reasoning approaches and offering insights into future research directions.

7.2 Future Directions

Despite significant progress in knowledge graph reasoning, several challenges remain unsolved. Current research primarily focuses on

reasoning within a single knowledge graph, often overlooking the potential of integrating knowledge from different languages and modalities. To address these gaps, we outline future directions that could advance the field.

The first future direction is reasoning on multi-modal knowledge graphs. These graphs combine structured knowledge with unstructured data such as images, videos, and audio. Reasoning on multi-modal knowledge graphs can uncover valuable insights that single-modal data cannot. For example, it can predict whether two images contain the same building or associate text descriptions with relevant multimedia content. Integrating multiple data types enhances the robustness and applicability of knowledge graph reasoning, making it possible to address more complex and diverse queries.

The second future direction is reasoning on cross-lingual knowledge graphs. Many knowledge graphs exhibit similar patterns despite being described in different languages. Mining these patterns can reveal useful information that transcends linguistic boundaries. Potential research directions include language-agnostic representation learning, multilingual logical rule extraction, and cross-lingual link prediction. These approaches aim to create models that understand and reason across different languages, enabling more inclusive and comprehensive knowledge graph applications. This can significantly enhance global information retrieval, allowing for more effective use of knowledge graphs in multilingual contexts.

REFERENCES

- [1] Ajith Abraham. 2005. Rule-Based expert systems. *Handbook of measuring system design* (2005).
- [2] A Acharya and S Adhikari. 2021. Alexa Conversations: An Extensible Data-driven Approach for Building Task-oriented Dialogue Systems. (2021).
- [3] Erik Arakelyan, Daniel Daza, Pasquale Minervini, and Michael Cochez. 2020. Complex query answering with neural link predictors. *arXiv preprint arXiv:2011.03459* (2020).
- [4] Erik Arakelyan, Pasquale Minervini, Daniel Daza, Michael Cochez, and Isabelle Augenstein. 2024. Adapting neural link predictors for data-efficient complex query answering. *Advances in Neural Information Processing Systems* 36 (2024).
- [5] Yushi Bai, Xin Lv, Juanzi Li, and Lei Hou. 2023. Answering Complex Logical Queries on Knowledge Graphs via Query Computation Tree Optimization. In *Proceedings of the 40th International Conference on Machine Learning*.
- [6] Antoine Bordes, Nicolas Usunier, Alberto Garcia-Duran, Jason Weston, and Oksana Yakhnenko. 2013. Translating embeddings for modeling multi-relational data. *Advances in neural information processing systems* 26 (2013).
- [7] C Buck and J Bulian. 2017. Ask the Right Questions: Active Question Reformulation with Reinforcement Learning. (2017).
- [8] Stefano Ceri, Georg Gottlob, Letizia Tanca, et al. 1989. What you always wanted to know about Datalog (and never dared to ask). *IEEE transactions on knowledge and data engineering* 1, 1 (1989), 146–166.
- [9] Xuelu Chen, Ziniu Hu, and Yizhou Sun. 2022. Fuzzy logic based logical query answering on knowledge graphs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 3939–3948.
- [10] William F Clocksin and Christopher S Mellish. 2003. *Programming in PROLOG*. Springer Science & Business Media.
- [11] William W Cohen. 2016. Tensorlog: A differentiable deductive database. *arXiv preprint arXiv:1605.06523* (2016).
- [12] R Das, S Dhuliawala, and M Zaheer. 2017. Go for a Walk and Arrive at the Answer: Reasoning Over Paths in Knowledge Bases using Reinforcement Learning.
- [13] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. 2024. From Local to Global: A Graph RAG Approach to Query-Focused Summarization. *arXiv preprint arXiv:2404.16130* (2024).
- [14] A Elghohary, D Peskov, and J Boyd-Graber. 2019. Can You Unpack That? Learning to Rewrite Questions-in-Context. Association for Computational Linguistics.
- [15] Katrin Erk. 2009. Representing Words as Regions in Vector Space. In *Proceedings of the Thirteenth Conference on Computational Natural Language Learning* (Boulder, Colorado) (CoNLL '09). Association for Computational Linguistics, USA, 57–65.
- [16] A Fader, L Zettlemoyer, and O Etzioni. 2014. Open Question Answering over Curated and Extracted Knowledge Bases (KDD '14). Association for Computing Machinery.
- [17] Luis Galárraga, Christina Teflioudi, Katja Hose, and Fabian M Suchanek. 2015. Fast rule mining in ontological knowledge bases with AMIE+. *The VLDB Journal* 24, 6 (2015), 707–730.
- [18] Luis Antonio Galárraga, Christina Teflioudi, Katja Hose, and Fabian Suchanek. 2013. AMIE: association rule mining under incomplete evidence in ontological knowledge bases. In *Proceedings of the 22nd international conference on World Wide Web*. 413–422.
- [19] Matt Gardner, Partha Talukdar, Jayant Krishnamurthy, and Tom Mitchell. 2014. Incorporating Vector Space Similarity in Random Walk Inference over Knowledge Bases. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Alessandro Moschitti, Bo Pang, and Walter Daelemans (Eds.). Association for Computational Linguistics, Doha, Qatar, 397–406. <https://doi.org/10.3115/v1/D14-1044>
- [20] D Guo, D Tang, N Duan, M Zhou, and J Yin. 2018. Dialog-to-Action: Conversational Question Answering Over a Large-Scale Knowledge Base. Curran Associates, Inc.
- [21] Will Hamilton, Payal Bajaj, Marinka Zitnik, Dan Jurafsky, and Jure Leskovec. 2018. Embedding logical queries on knowledge graphs. *Advances in neural information processing systems* 31 (2018).
- [22] Shizhu He, Kang Liu, Guoliang Ji, and Jun Zhao. 2015. Learning to Represent Knowledge Graphs with Gaussian Embedding. In *Proceedings of the 24th ACM International Conference on Information and Knowledge Management* (Melbourne, Australia) (CIKM '15). Association for Computing Machinery, New York, NY, USA, 623–632.
- [23] Vinh Thinh Ho, Daria Stepanova, Mohamed H Gad-Elrab, Evgeny Kharlamov, and Gerhard Weikum. 2018. Rule learning from knowledge graphs guided by embedding models. In *The Semantic Web—ISWC 2018: 17th International Semantic Web Conference*. Springer, 72–90.
- [24] E Kacupaj, J Plepi, K Singh, and H Thakkar. [n. d.]. Conversational Question Answering over Knowledge Graphs with Transformer and Graph Attention Networks. Association for Computational Linguistics.
- [25] Endri Kacupaj, Kuldeep Singh, Maria Maleshkova, and Jens Lehmann. 2022. Contrastive Representation Learning for Conversational Question Answering over Knowledge Graphs. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management (CIKM '22)*. Association for Computing Machinery, New York, NY, USA.
- [26] Magdalena Kaiser, Rishiraj Saha Roy, and Gerhard Weikum. 2021. Reinforcement learning from reformulations in conversational question answering over knowledge graphs. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval*. 459–469.
- [27] Jiho Kim, Yeonsu Kwon, Yohan Jo, and Edward Choi. 2023. KG-GPT: A general framework for reasoning on knowledge graphs using large language models. *arXiv preprint arXiv:2310.11220* (2023).
- [28] Ni Lao, Tom Mitchell, and William Cohen. 2011. Random walk inference and learning in a large scale knowledge base. In *Proceedings of the 2011 conference on empirical methods in natural language processing*. 529–539.
- [29] Leonid Libkin. 2004. *Elements of finite model theory*. Vol. 41. Springer.
- [30] X Lin and R Socher. [n. d.]. Multi-Hop Knowledge Graph Reasoning with Reward Shaping. In *EMNLP 2018*.
- [31] Yankai Lin, Zhiyuan Liu, Maosong Sun, Yang Liu, and Xuan Zhu. 2015. Learning entity and relation embeddings for knowledge graph completion. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 29.
- [32] Lihui Liu, Yuzhong Chen, Mahashweta Das, Hao Yang, and Hanghang Tong. 2023. Knowledge Graph Question Answering with Ambiguous Query. In *Proceedings of the ACM Web Conference 2023*.
- [33] Lihui Liu, Boxin Du, Heng Ji, ChengXiang Zhai, and Hanghang Tong. 2021. Neural-Answering Logical Queries on Knowledge Graphs. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '21)*.
- [34] Lihui Liu, Boxin Du, Jiejun Xu, Yinglong Xia, and Hanghang Tong. 2022. Joint Knowledge Graph Completion and Question Answering. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (Washington DC, USA) (KDD '22). Association for Computing Machinery, New York, NY, USA, 1098–1108.
- [35] Lihui Liu, Blaine Hill, Boxin Du, Fei Wang, and Hanghang Tong. 2024. Conversational Question Answering with Language Models Generated Reformulations over Knowledge Graph. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 839–850. <https://doi.org/10.18653/v1/2024.findings-acl.48>
- [36] Lihui Liu, Zihao Wang, Ruizhong Qiu, Yikun Ban, Eunice Chan, Yangqiu Song, Jingrui He, and Hanghang Tong. 2024. Logic Query of Thoughts: Guiding Large Language Models to Answer Complex Logic Queries with Knowledge Graphs. *arXiv:2404.04264* [cs.LG] <https://arxiv.org/abs/2404.04264>
- [37] David Marker. 2006. *Model theory: an introduction*. Vol. 217. Springer Science & Business Media.